Text REtrieval Conference (TREC)

...to encourage research in information retrieval from large text collections.

TREC 2024 Tip-of-the-Tongue (ToT) Track

Information Retrieval using LLM (RAG)

Presenter: Shuai Yang
Advisor: Prof. Jeremy Blackburn, Prof. Sujoy Sikdar

Presented on Oct 7th, 2024

Tip of the Tongue (ToT) Task - Definition

An **identification task** where the searcher has previously experienced or consumed an item but **cannot recall a reliable identifier**.(i.e., “It’s on the tip of my tongue…”)



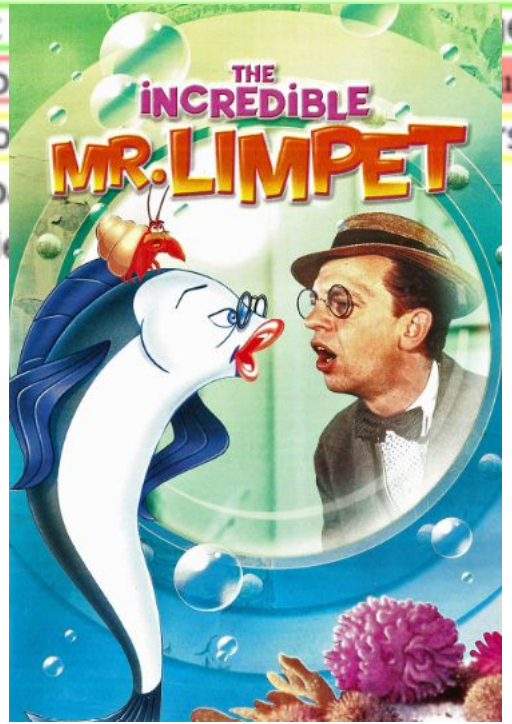
ToT information requests (or “queries”) are **verbose** and contain a wide-range of **complex phenomena**.

- long-term memory
- mentions of uncertainty
- false memories

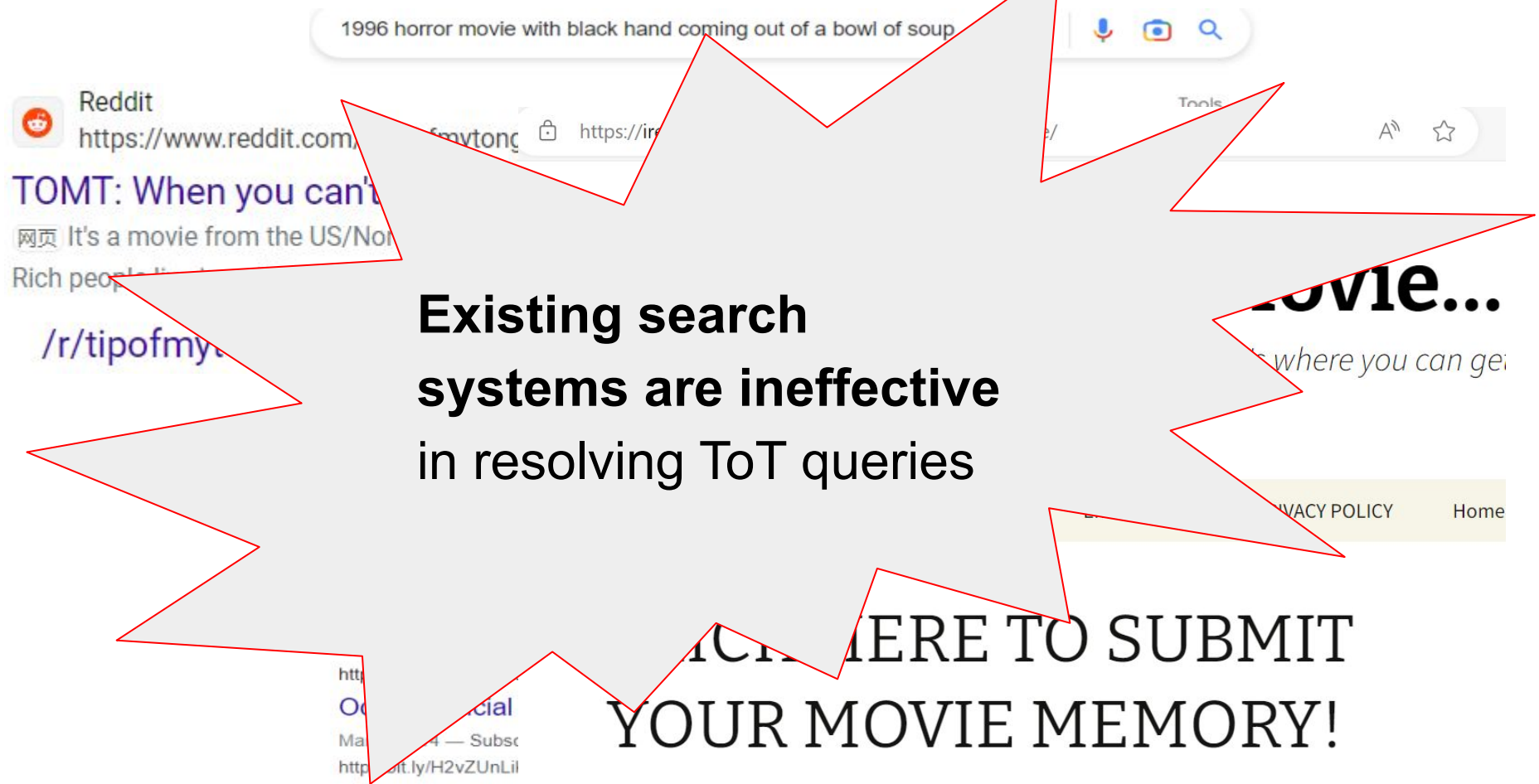
Tip of the Tongue (ToT) Task - Movie Example

I couldn't have been older than 4, so this was around 2002. I watched a movie with my parents (or so I thought) and despite never watching it again, it became my favorite. It ed plot man who went on some kind of adventure and turned into a fish. I also ncertainty school of some sort? It seemed like a slightly old movie, but it was in co 's visual and changed to animation. For weeks after I saw this movie I told my p insisted it was a dream so I let it go. Does anyone know what this movi

Correct answer: The Incredible Mr. Limpet

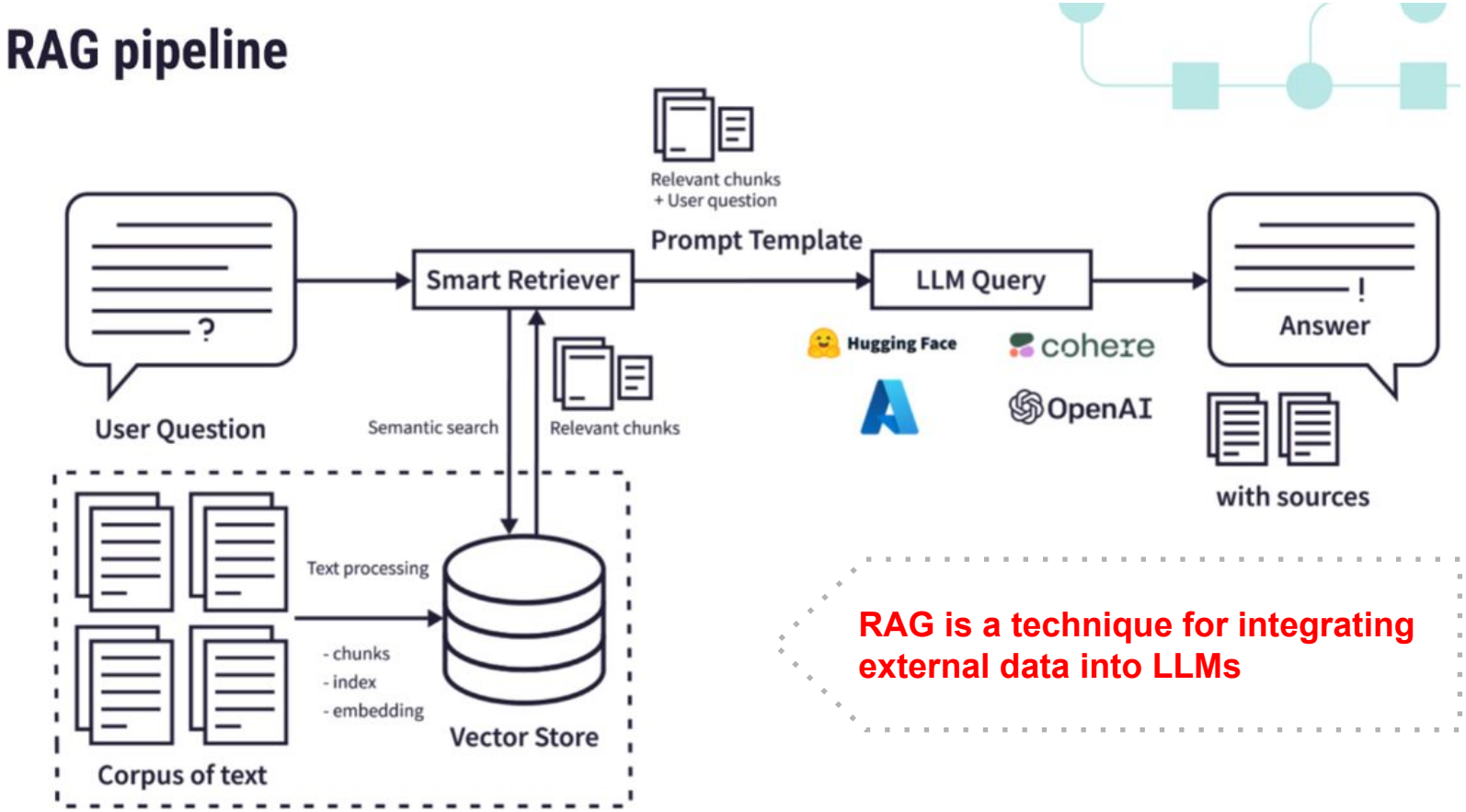


Tip of the Tongue (ToT) Task - Motivation



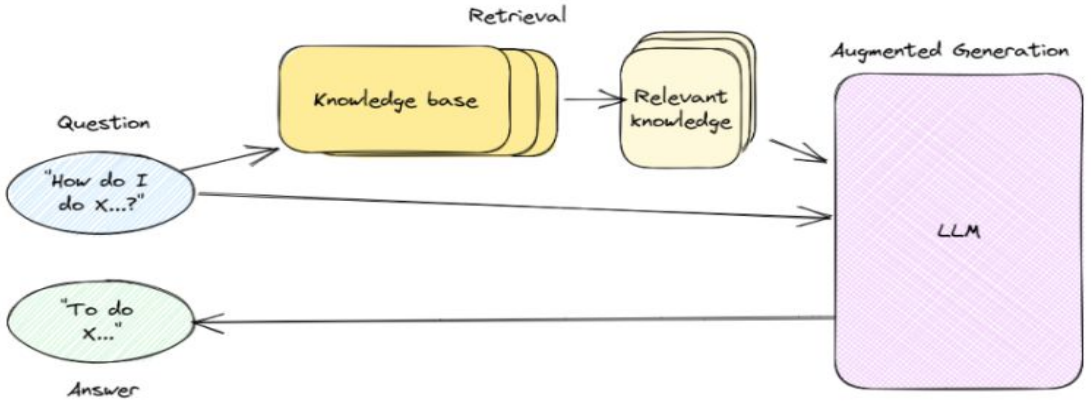
Existing search systems are ineffective in resolving ToT queries

Retrieval-Augmented Generation (RAG)

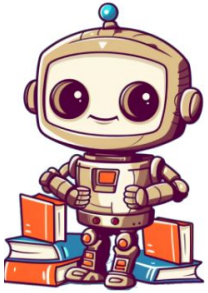


Why RAG is powerful for solving IR problems

- Up-to-Date Information



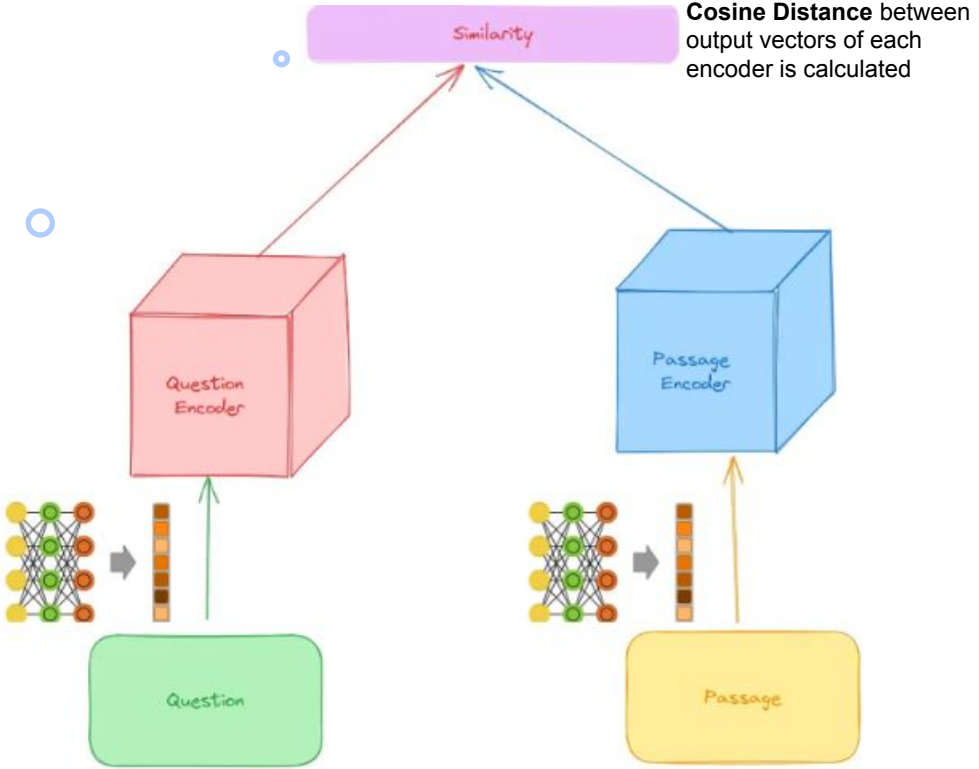
- Contextual Relevance



Retrieval Technique - Dense Passage Retrieval (DPR)

DPR is a technique used in NLP to efficiently retrieve relevant **passages** of text in response to a **query**.

bi-encoder architecture



Evaluate Metrics on Submitted Runs



- Discounted Cumulative Gain (DCG)
- Reciprocal Rank (RR)
- Recall

2023 Submissions

Run ID	Group ID	Attestation	External Data	Re-rank	NDCG@10	NDCG@1000	MRR@1000	Recall@1000
dpr-1000-rerank-robin	CMU-LTI		✓		**0.5169**	**0.5554**	**0.5016**	0.7933
dpr-100-rerank	CMU-LTI		✓		0.4631	0.5070	0.4523	0.8000
pre_aug_vat_max4	snuldilab	✓	✓		0.2471	0.3301	0.2263	0.8467
pre_aug_vat_max4_origin	snuldilab	✓	✓		0.2352	0.3224	0.2138	**0.8533**
pre_aug_vat	snuldilab	✓			0.2398	0.3206	0.2204	0.8200
webis-t53b-01	Webis				0.2404	0.2894	0.2171	0.6267
baseline_gpt4_db*	Baseline	✓			0.2495	0.2624	0.2314	0.3733
dpr-abstract-1000-robin	CMU-LTI		✓		0.2479	0.2584	0.2367	0.3600
WatS-TDR	UWaterlooMDS	✓			0.1635	0.2480	0.1515	0.7533
webis-t5-f	Webis				0.1833	0.2459	0.1698	0.6267
ufmgG4mBQD	ufmg	✓		✓	0.2304	0.2404	0.2002	0.3733
webis-fus-01	Webis				0.1450	0.2092	0.1318	0.6267
webis-t5-01	Webis				0.1331	0.2090	0.1254	0.6267
ufmgDBmBQ	ufmg	✓		✓	0.1813	0.2090	0.1636	0.3933
WatS-DR	UWaterlooMDS	✓			0.1310	0.2043	0.1195	0.7067
WIS_LSR_UNICOIL	WIS_TUD	✓	✓		0.1304	0.2042	0.1241	0.6400
ufmgDBmBQD	ufmg	✓		✓	0.1647	0.1998	0.1507	0.4067
WIS_LSR_SPLADE_ASM_QMLP	WIS_TUD	✓	✓		0.1083	0.1843	0.1018	0.6400
webis-bm25r-1	Webis		✓		0.1137	0.1672	0.1029	0.5200
ufmgG4dTQD	ufmg	✓		✓	0.1214	0.1668	0.1189	0.3733
dpr-abstract-100-rerank	CMU-LTI		✓		0.1235	0.1532	0.1219	0.3600
endicott_unc_boost_pred	endicott-unc	✓			0.1089	0.1516	0.0954	0.4533
endicott_unc_boost_oracle	endicott-unc	✓			0.1018	0.1439	0.0907	0.4267
baseline_distilbert*	Baseline	✓			0.0820	0.1426	0.0683	0.5467
baseline_bm25*	Baseline	✓			0.0930	0.1388	0.0837	0.4467
dpr_multidoc_roberta	CIIR				0.0641	0.1248	0.0634	0.4867
WatS-TDR-RR	UWaterlooMDS				0.0362	0.1244	0.0344	0.7533
endicott_unc_baseline	endicott-unc	✓			0.0749	0.1116	0.0663	0.3667
endicott_unc_boost_conf	endicott-unc	✓			0.0749	0.1116	0.0663	0.3667
ufmgDBmBdTQD	ufmg	✓		✓	0.0566	0.1108	0.0505	0.4067
WIS_DB_FT	WIS_TUD	✓	✓		0.0000	0.0720	0.0010	0.6933
RSLTOTY	RSLTOT	✓	✓		0.0169	0.0565	0.0128	0.3267
runid1	WaterlooClarke	✓			0.0095	0.0095	0.0061	0.0200

Table 2: Summary of results. Best scores are in bold. The baselines are marked with an asterisk.

Future Work on ToT Task

- Re-ranking 

- External data 

- Fine-tuning LLM 

    - Training my own Dense Passage Retrieval (DPR) Model 

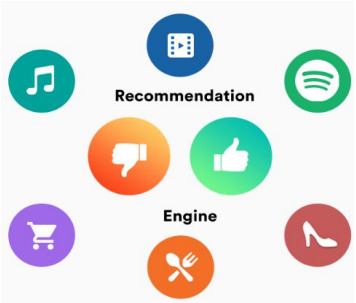
Tentative Future Projects



- Generative Information Retrieval
- Retrieval-enhanced LLM
- Multimodal RAG
- and their Applications in Social Media/Health/Recommendations/Finance, etc









Thank you!